

A Systematic Experimentation: How does sugar environment affect agent sugar accumulation and wealth inequality in SugarScape 2 and 3?

Word Count: 1498

Declaration: ChatGPT has been used to fix coding errors as well as tidy some writing mistakes.

Aim

Epstein and Axtell (1996) presented a series of socioeconomic agent-based models simulating artificial societies on Sugarscape. Through adapting and modifying parameters to the agents and environment, this will allow experimentation of more complex behavioral interactions.

This experiment compares Sugarscape 2 (S2) and Sugarscape 3 (S3) under varying sugar environments. This includes one peak, two peak, and four peak environments, representing a spatial distribution gradient from highly concentrated to more evenly distributed resources. Key measuring metrics recorded mean wealth, median wealth and Gini index.

The aim is to investigate whether spatial sugar distribution affects wealth accumulation and inequality. This does not only compare across environments but also between a survival-based model where wealth accumulates among survivors (S2), and a replacement ruled model where agent replacements may stabilise but not eliminate inequality (S3).

Methods

The experiment used S2 and S3 derived from NetLogo Models Library. In both models, agents are born with a heterogeneous genetic endowment of vision and metabolism, which collectively shape how they behave in Sugarscape. Agents move to the highest-sugar patch within their vision and collect it, decrementing sugar through metabolism

each tick. In S2, agents survival depends on maintaining a sugar wealth level above 0, where population can decline over time. S3 adopts a similar movement and consumption mechanism but with an additional replacement rule. This replaces agents upon death, renewing population. Results therefore measure the wealth of agents active at each tick.

Model Adaptations

The model is adapted to simulate three different sugar environments. The one peak represents a highly concentrated resource hill. Two peaks as a baseline for comparison, designed to closely replicate original S2 model. The four peak is distributed resources into 4 concentrated hills across the grid. This adaptation experiment systematically whether fewer concentration locations increases competition and inequality, while distributing sugar across more locations improves accessibility. As shown in Figure 1, the environmental setup moves from concentrated to a more equally distribution.

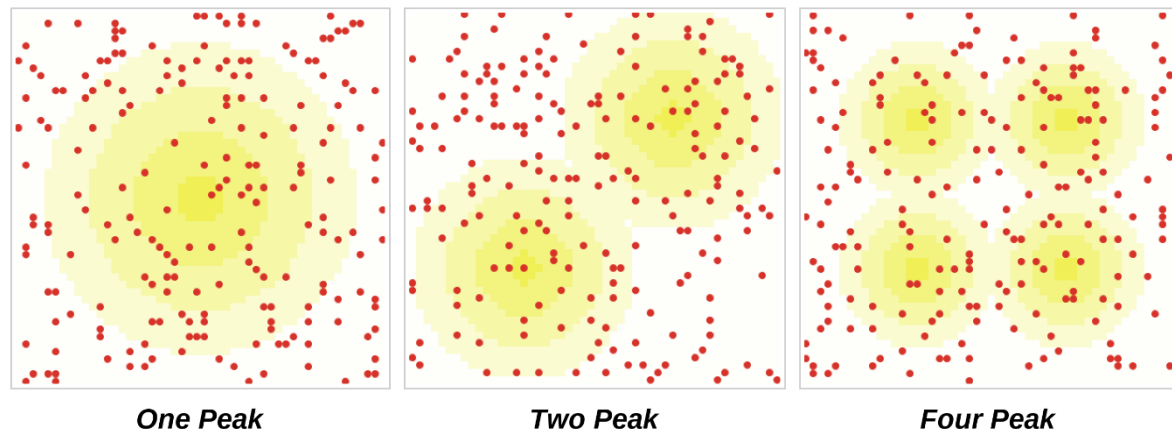


Figure 1. Sugar Environment Setup

Table 1. Hypothesis

One Peak	Environment cause higher inequality due to more concentrated access to patches with highest capacity
Two Peaks	As the base line environment, environment has moderate wealth accumulation and inequality.
Four Peaks	Environments improve accessibility, thereby reduces inequality

Maximum patch capacity was kept at 4 across all environments with a constant growback rule. Total sugar capacity was kept approximately constant across the three environments, with a difference of only less than 1%. Although the environments weren't perfectly identical, it is small enough that differences in wealth inequality observed are most likely to reflect how sugar is spatially arranged rather than differences in the total capacity of sugar.

Table 2. Sugar Capacity by Environment Type

Environment Type	Environment Sugar Capacity	Maximum Patch Sugar Capacity
One Peak	2376	4
Two Peak	2365	4
Two Peak	2365	4

BehaviorSpace Experiment Design

BehaviorSpace was used to run systematic experiments across both models with the three environments. Each condition was simulated on 30 repetitions over 500 ticks, with metrics recorded every 10 ticks and at final tick. Initial population was kept at 300 in all runs, allowing both dynamic change and simulation variation to be compared across conditions.

Model Metrics

Several reporters were added to measure model behaviour. The main metrics analysed were mean-wealth, median-wealth, and gini-index. Mean wealth captures average wealth, median wealth for central tendency, and Gini index for inequality, where 0 indicates equality and 1 for inequality. Reporters were calculated over the current agents at each tick in both models. Additional reporters, including average sugar per patch and top 10% wealth were monitored.

Results

Mean Wealth

Mean wealth in S2 experienced a steady gradient increasing over time across all environments, with four peaks producing the highest accumulation and one peak the lowest. The widening gap suggests environment does not affect initial access to resources but creates cumulative differences in long-run accumulation as shown in Figure 2 and Table 3.

Mean wealth in S3 was much lower, with minor early fluctuations before reaching relatively flat. Differences between environments exists, with four peaks highest and one peak lowest. Table 3 shows final values of 30.6, 31.4 and 33.0 for one, two and four peaks, compared with 467.4, 552.7 and 575.0 in S2. This suggests that environmental concentration shapes wealth accumulation in both models, but outcomes also depend on survival and replacement rules that may limit accumulation over time.

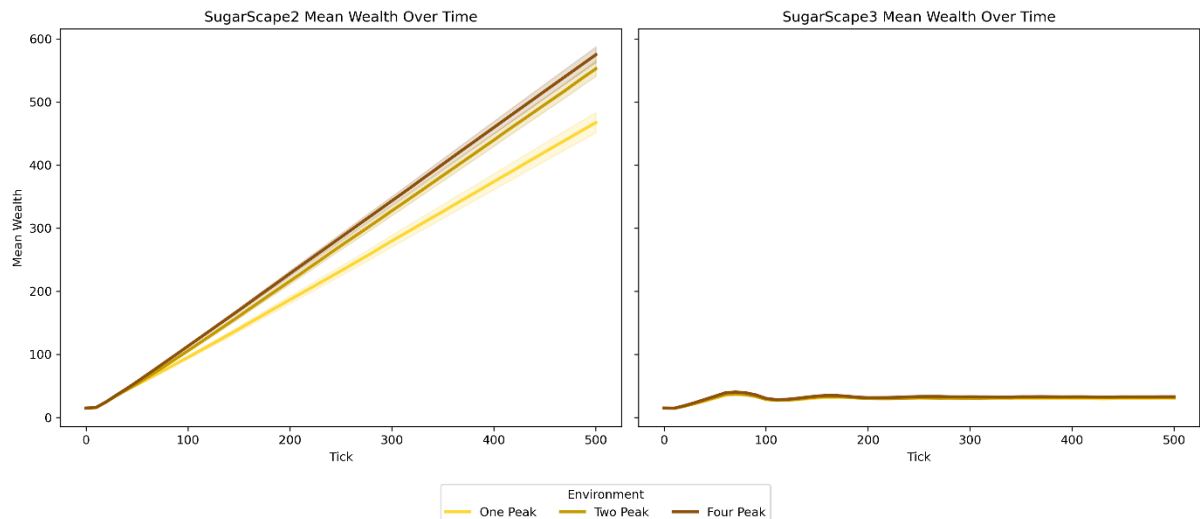


Figure 2. Mean Wealth Over Time, Sugarscape 2 and 3

Gini Coefficient

The results have showed that wealth gaps appeared at early ticks in both models. This suggests a proportion of agents were able to reach high capacity patches where patterns stabilise or fluctuates differently in different environment types.

S2 showed different levels of inequalities throughout simulation. One peak found highest inequality followed by two peaks and four peaks, with final Gini values of 0.471, 0.353 and 0.312 respectively. When resources are concentrated in fewer locations, only a small number of agents can access higher capacity patches, driving greater wealth inequality.

Inequality remained high in S3 but differences between environments were smaller and more stable with final Gini values of 0.480, 0.468 and 0.453 respectively. Although following the same pattern, the gaps between environment types were smaller in S2, as constant replacement of agents in S3 tries preventing inequality from accumulating over time.

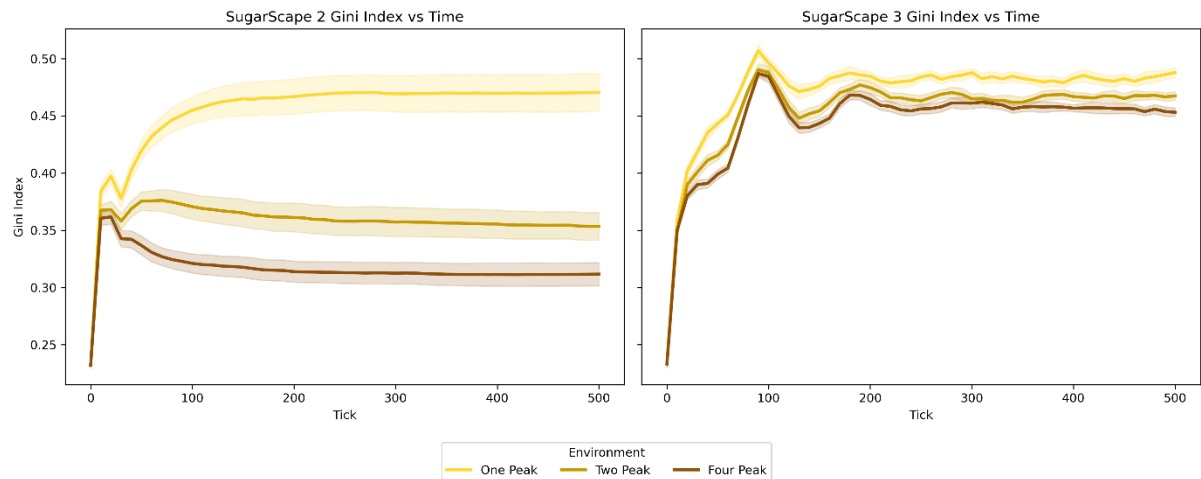


Figure 3. Gini Index Over Time, Sugarscape 2 and 3

Final Mean and Median Wealth

Comparing mean and median values in wealth shows distribution of wealth within the simulation. Mean wealth exceeds median wealth across all environment types in S3, indicating that even under replacement rule, a more well-off population exists and is likely to skew the average of agent wealth. The situation is mixed in S2, with mean above median in two and four peaks, and one peak below median. These indicated uneven distributions of wealth across different environments, where wealth is high and inequality exists.

Table 3. Summary Statistics (Final Tick)

Model	Environment	Mean Wealth	Median Wealth	Gini Index
S2	one-peak	467.4	473.0	0.471
S2	two-peak	552.7	497.5	0.353
S2	four-peak	575.0	504.4	0.312
S3	one-peak	30.6	19.5	0.488
S3	two-peak	31.4	20.9	0.468
S3	four-peak	33.0	22.2	0.453

Final Gini Coefficient

The final Gini boxplots in Figure 4 show that one peak produced the highest final inequality in both models, while four peaks produced the lowest. In Figure 4, S2 showed more diverse levels of Gini index across repetitions, especially in the one-peak environment, indicating less stable and higher inequality. S3 showed less variation yet positioned at high Gini levels.

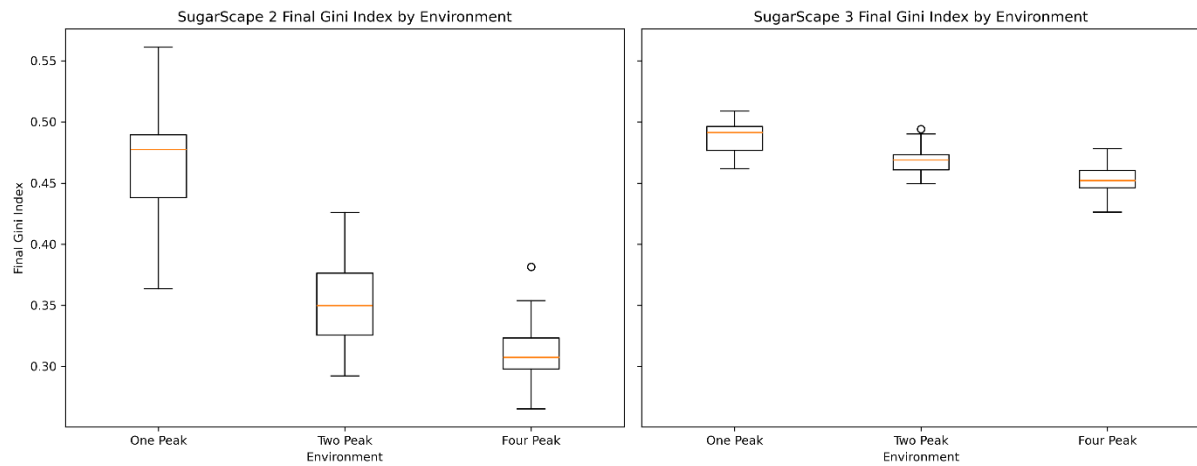


Figure 4. Final Gini Index by Environment Type

Discussion

The results have showed that local interactions of heterogeneous agents in varied spatial structures impacts macro-levels of how much wealth agents accumulate and wealth inequality (Tesfatsion, 2002). In both models, resource concentration increases inequality. When sugar is concentrated, agents compete for limited number of high-capacity patches, limiting access to minority. Distributing sugar across four peaks improves patch accessibility, wealth outcomes and reduce inequality.

These interactions were found stronger in S2, where agents continue accumulating sugar, where less fit agents are removed from the simulation. Agents who are genetically and physically advantageous at initial are more likely to reach high sugar patches, thereby able to keep accumulating wealth. However, results should not be interpreted as the population gets wealthier collectively. As deaths are not replaced in S2, average wealth reflects only those who are active. This explains why wealth is higher in S2 and gaps increases over time, where it is not only about accessibility, but also agent's ability to survive and keep accumulating.

S3 behaves differently due to an additional replacement rule with new, low-wealth agents. Mean and median values are relatively lower compared to S2 due to constant replacement. However, replacement does not remove inequality as access to high-capacity patches remains the same. Therefore, low mean wealth and high Gini values showed that wealth does not accumulate to higher levels while remaining unequal.

Comparing the two models shows that spatial distribution shapes agents wealth accumulation to an extent, yet behavioural rules determine whether wealth accumulation grows over time. In both models, concentrated resources produced

higher inequality, indicating that sugar distribution is indeed a key driver of inequality in Sugarscape. Yet, this experiment posed several limitations, such as different movement or metabolism rules were not incorporated. Moreover, keeping sugar capacity to be similar but not perfectly identical across environments may create larger clustering of the higher capacity patches in the centre of the grid. This may also lead to results that reflects concentration effects. Therefore, extending the experiment would give a more holistic understanding of how environmental and behavioural factors shape inequality collectively.

Conclusion

To conclude, the sugar environment does affect agent wealth accumulation and inequality in both models. The results show that concentrated environments increase inequality, while more evenly spatially distributed environments improve wealth outcomes especially in S2. Across both models, one peak is often recognised with highest Gini index and four peak with the lowest.

The effect on wealth accumulation was comparably stronger in S2, where agents could accumulate wealth over time. In S3, agent replacement rules limited accumulation, causing low overall wealth levels and relatively high inequality across all environments. This highlights that inequality is shaped not only by resource distribution, but agent's ability to accumulate wealth over time.

Future research could examine how varying initial population sizes interact with environment types and extending with resource decay rules. To achieve a more holistic economic environment, future research can also consider an "overlapping generations" economy, where new agents are able to inherit wealth from ancestor agents rather than entering the world with random characteristics (Teshfatsion, 2002).

References

Epstein, J.M. and Axtell, R.L. (1996) 'Life and Death on the Sugarscape', *Growing Artificial Societies: Social Science from the Bottom Up*. Available at: <https://doi.org/10.7551/mitpress/3374.001.0001>.

Tesfatsion, L. (2002) 'Agent-Based Computational Economics: Growing Economies From the Bottom Up'. Available at: <https://doi.org/10.1162/106454602753694765>.

Appendix

The experiment used BehaviorSpace to record the main metrics for each simulation run. Reporters were collected at intervals of 10 ticks during the simulation as well as at the final tick. The code used as reporters are as below.

```
;;  
;; Add Reporters  
;;  
  
; overall wealth level of the population  
to-report mean-wealth  
  if not any? turtles [ report 0 ]  
  report mean [sugar] of turtles  
end  
  
; wealth between poorest to richest, take the median value to inspect  
to-report median-wealth  
  if not any? turtles [ report 0 ]  
  report median [sugar] of turtles  
end  
  
; total capacity of sugar patches in the environment  
to-report environment-capacity  
  report sum [max-psugar] of patches  
end  
  
; maximum sugar capacity per patch  
to-report max-patch-capacity  
  report max [max-psugar] of patches  
end  
  
; gini index for inequality measure  
to-report gini-index  
  if not any? turtles [ report 0 ]  
  if sum [sugar] of turtles = 0 [ report 0 ]  
  report (gini-index-reserve / count turtles) / 0.5  
end  
  
; wealth percentage of the top 10% of the population  
to-report top10-wealth  
  if not any? turtles [ report 0 ]  
  let wealths sort [sugar] of turtles  
  let n count turtles  
  let cutoff max list 1 round (0.1 * n)  
  let richest sublist wealths (n - cutoff) n  
  if sum wealths = 0 [ report 0 ]  
  report sum richest / sum wealths  
end  
  
; average per sugar patch  
to-report mean-patch-sugar  
  report mean [psugar] of patches  
end
```